

EX.NO:7 IMPLEMENTATION OF A MULTI-THREAD PROGRAM

Program:

```
import java.util.Arrays;
import java.util.Random;

class ArrayThread {
    static int[] arr = new int[10];

    static class Generate extends Thread {
        public void run() {
            Random r = new Random();
            for (int i = 0; i < arr.length; i++) {
                arr[i] = r.nextInt(100);
            }
            System.out.println("Original Array: " + Arrays.toString(arr));
        }
    }

    static class Ascending extends Thread {
        public void run() {
            int[] temp = arr.clone();
            Arrays.sort(temp);
            System.out.println("Ascending Order: " + Arrays.toString(temp));
        }
    }

    static class Descending extends Thread {
        public void run() {
            int[] temp = arr.clone();
            Arrays.sort(temp);
            // Reverse for descending
            for (int i = 0; i < temp.length / 2; i++) {
                int t = temp[i];
                temp[i] = temp[temp.length - 1 - i];
                temp[temp.length - 1 - i] = t;
            }
        }
    }
}
```

```

        System.out.println("Descending Order: " + Arrays.toString(temp));
    }
}

public static void main(String[] args) throws Exception {
    Generate t1 = new Generate();
    Ascending t2 = new Ascending();
    Descending t3 = new Descending();

    t1.start();
    t1.join(); // Wait for generation to complete

    t2.start();
    t3.start();

    t2.join();
    t3.join();

    System.out.println("All threads completed.");
}
}

```

Output:

Original Array: [42, 15, 88, 4, 73, 29, 91, 10, 56, 34]
 Ascending Order: [4, 10, 15, 29, 34, 42, 56, 73, 88, 91]
 Descending Order: [91, 88, 73, 56, 42, 34, 29, 15, 10, 4]
 All threads completed.